

Tharindu Dhanushka

AI / ML Engineer · Computer Engineering Fresh graduate · University of Ruhuna

tharindubandara126@gmail.com · +94 75 392 9283 · [GitHub](#) · [LinkedIn](#) · [Omdena Profile](#) · Colombo, Sri Lanka

SUMMARY

Computer Engineering Fresh Graduate with hands-on production experience in ML deployment, model fine-tuning, multi-agent AI systems, and statistical analysis. Ranked 778th globally (55th in Sri Lanka) at IEEEExtreme 18.0 out of 8,700+ teams. Finalist in the Veracity Group Gen AI Hackathon (top 15 of 70+ teams). Specialised in Python-based ML pipelines and agentic LLM systems from data ingestion and feature engineering to model serving with a track record of shipping applications in real-world environments. Conducting active biomedical research applying multimodal ML to physiological signal data.

SKILLS

ML & AI: TensorFlow, Scikit-learn, PyTorch, QLoRA fine-tuning, LSTM, Random Forest, CNN (ResNet50), LangChain, CrewAI, RAG, Agentic AI

LLM & NLP: OpenAI, Google Gemini, GroqLM, HuggingFace, Cosine Similarity, multi-agent orchestration, Dialogflow CX

Data & Statistics: Pandas, NumPy, Matplotlib, Seaborn, PCA, statistical modelling, time-series analysis

Languages: Python, JavaScript, SQL, C, C++

Backend: Django, Flask, Node.js, Next.js, REST APIs

Frontend: React, Tailwind CSS, Streamlit, Gradio

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Vector Databases (Qdrant, Chroma)

DevOps & Tools: Docker, CI/CD, Git, Jenkins, Linux, Jira, Agile, Postman, AssemblyAI

EXPERIENCE

AI / ML Engineer · [OXZON AI](#) · Internship

May – Nov 2025 · Sri Lanka

- Shipped 2 production ML applications, reducing average inference latency by refactoring model - serving architecture and integrating async API endpoints with Flask and Docker.
- Raised LLM task accuracy from **71%** → **89%** on a domain classification benchmark by fine-tuning a pre-trained model using QLoRA on a curated 4,000 sample internal dataset.

Junior ML Engineer · [Omdena Nigeria Chapter](#) · Collaborator

Jun – Aug 2024 · Remote

- Improved time-series forecast accuracy by 18% over a linear baseline by building and ensembling LSTM and Random Forest Regressor models, then deploying the pipeline via a Flask API.
- Delivered a collaborative ML environment serving 10+ contributors, cutting model iteration cycles from days to hours by containerising the inference service with Docker.

Junior Data Scientist · [Omdena Zambia Chapter](#) · Collaborator

May – Jun 2024 · Remote

- Deployed a production-ready ML application, reducing manual data-processing time by designing an end-to-end pipeline from multi-source data ingestion to model inference.
- Improved dataset quality by sourcing and cleaning structured data from 3+ external sources, increasing usable training samples.

PROJECTS & RESEARCH

[SLM Fine-Tuning with RL Optimisations](#) - Python · QLoRA · GRPO · PPO

- Fine-tuned a small language model using QLoRA with GRPO and PPO reward-based optimisation, implementing custom reward functions and evaluating alignment quality across multiple checkpoints.
- Validated training stability and alignment improvement across all checkpoint iterations a graduate-level applied ML research skill rarely seen at undergraduate level.

[MeetingMinute - Multi-Agent Meeting Summariser](#) - CrewAI · OpenAI · AssemblyAI · MongoDB · Qdrant · Composio

- Built a real-world meeting summariser using a **CrewAI multi-agent system (4 crews, 8 agents)** in sequential processing, integrating AssemblyAI for speech transcription and a Text-to-Speech API for audio output.
- Integrated OpenAI as the LLM backbone, Qdrant as the vector database for semantic retrieval, MongoDB for persistent storage, and Composio API for automated email triggering on meeting completion.
- Recognised as a finalist in the Veracity Group Gen AI Hackathon top 15 of 70+ teams in a 1-week competition.

Multilingual AI Chatbot with Persistent Memory - LangChain · Gemini · MongoDB · Gradio · Hugging Face

- Deployed a production chatbot supporting 5 languages with cross-session memory persistence via MongoDB, live on Hugging Face Spaces reducing user re-prompting effort.
- Integrated Google Gemini AI for generation and LangChain for orchestration, achieving sub-2s response latency across all supported language pairs.

RAG-Powered PDF Research Tools - GroqLM · LangChain · HuggingFace · Chroma · Qdrant · Gradio

- Built [PDF Research Assistant](#) (GroqLM + LangChain) with vector-based retrieval and conversation memory for advanced document question-answering, deployed via Gradio.
- Built [QBot](#), a PDF query chatbot using LangChain, HuggingFace embeddings, and Chroma vector store for context-aware retrieval demonstrating end-to-end RAG pipeline development.

Chest Disease Detection & Classification - CNN · ResNet50 · TensorFlow

- Achieved 94% classification accuracy on chest X-ray data by fine-tuning ResNet50 with transfer learning, applying data augmentation to address class imbalance across 4 disease categories.

Biomedical ML Research · Active - Python · TensorFlow · Statistical Modelling · Ultrasound Segmentation

- Investigating cardiovascular signal correlates (HRV, blood pressure dynamics) as predictors of skeletal muscle volume across 8 muscle groups using ultrasound imaging and ensemble ML targeting Elsevier / Springer publication.
- Building a multimodal analysis pipeline with demographic, nutritional, and psychophysiological features, combining public datasets with manually annotated ultrasound data using a custom RST-UNet segmentation model.

EDUCATION

BSc (Hons) Computer Engineering · [University of Ruhuna](#) Mar 2022 – Present · Galle, Sri Lanka

- Relevant coursework: Artificial Intelligence, Machine Learning, DevOps Engineering, Computer Architecture, Data Structures & Algorithms, Design Patterns, OOP, Databases, Embedded Systems.

COMPETITIONS & HACKATHONS

IEEEExtreme 18.0 Programming Competition

Nov 2024

- Ranked **778th globally and 55th in Sri Lanka** out of 8,700+ teams worldwide.

Veracity Group Gen AI Hackathon

2024

- Finalist - **top 15 of 70+ teams** in a 1-week competition. Built MeetingMinute, a real-world multi-agent AI summariser using CrewAI.

CERTIFICATIONS

- [TensorFlow Developer Professional Certificate](#) · DeepLearning.AI · Coursera · May 2024
- [Machine Learning with Python](#) · IBM · Coursera · May 2024
- [Neural Networks and Deep Learning](#) · DeepLearning.AI · 2024
- [Building Agentic RAG with LlamaIndex](#) · DeepLearning.AI · 2024
- [Introducing Multimodal Llama 3.2](#) · DeepLearning.AI · 2024
- [Building Your Own Database Agent](#) · DeepLearning.AI · 2024

REFERENCES

Available on request ·

Dr. Rajitha Udawalpola (HOD, EIE, University of Ruhuna · rajitha@eie.ruh.ac.lk) ·

Dr. Kushan Sudheera (Senior Lecturer, EIE, University of Ruhuna · kushan@eie.ruh.ac.lk)